

The Polylogarithm–Polygamma Bridge: An Exact Finite Fourier Transform, Clausen-Type Laws at Rational Arguments, and the Layers Below the Digamma

Vladimir Reshetnikov*

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Abstract

The classical cyclotomic formula expressing Li_n at a q -th root of unity as a finite sum of polygamma values is one half of an *exact finite Fourier transform over $\mathbb{Z}/q$* ; we record the inverse direction and its structural consequences: the polygamma relation lattice (reflection + multiplication) and the cyclotomic-polylogarithm relation lattice (inversion/reflection + distribution) are DFT-conjugate — which explains their identical dimension counts $\varphi(q)/2$ per denominator grid — and Catalan’s constant is exactly the $q = 4$ odd component of the parity split. Pushing the bridge into explicit “store-level” laws unifies the fifteen closed forms of MathWorld’s *Polygamma Function* special-cases block into instances of sine/cosine-weighted Clausen-DFT laws valid at *every* rational argument; the trigamma law turns out to be classical (Lewin 1991), and the corpus now carries a 54-entry verified store of $\psi^{(m)}(p/q)$ closed forms ($m \leq 3$, $q \in \{3, 4, 5, 6, 8, 12\}$) in canonical Clausen atoms. A genuine sharpening emerged en route: the folklore collapse $\text{Cl}_3(\text{rational angle}) \in \mathbb{Q} \cdot \zeta(3)$ holds exactly for denominators with $\varphi(q) \leq 2$ (the crystallographic angles), and fails first at $q = 5, 8, 12$, where $\text{Cl}_3(2\pi/5)$, $\text{Cl}_3(\pi/4)$, $\text{Cl}_3(\pi/6)$ are genuine new atoms with exact $L(3, \chi)$ bridges. Below the digamma, integrating Kummer’s series shows that $\psi^{(-2)}$ at rationals lands in the weight-2 Clausen lattice with $\zeta'(-1)$ as the one new constant; Binet’s second formula and the Malmsten–Vardi log-log integrals join the same picture as the Lambert-kernel and Mellin faces of one bridge; and the generalized-Stieltjes layer splits by character parity — odd combinations of $\gamma_1(k/q)$ reduce classically (Malmsten 1846), while even combinations are elementary-equivalent to Deninger’s $\zeta''(0, \cdot)$ layer (Blagouchine 2015) and resist all standard atoms. Everything is verified numerically at 40–100 digits, most of it dual-engine, with PSLQ-canary discipline; epistemic status is stated per result.

1 Introduction: one bridge, many faces

Polylogarithms at roots of unity and polygamma functions at rational arguments are two spellings of the same finite-dimensional object. This article assembles the recorded results of a research program that made the dictionary exact and then walked it in both directions and *downward* — through $\log \Gamma$ into negative polygamma order, Malmsten’s integrals, and the generalized Stieltjes constants. The organizing picture is a ladder of bridges:

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object	polylogarithm side (at roots of unity)	new atom
$\psi^{(m)}, m \geq 1$	$\text{Re} / \text{Im Li}_{m+1}$ — exact two-way DFT	—
$\log \Gamma = \psi^{(-1)}$	Li_1 (Kummer’s log-sine series)	$\log 2\pi$
$\psi^{(-2)}$	$\text{Cl}_2 = \text{Im Li}_2$ (§5)	$\zeta'(-1)$
$\psi^{(-3)}$	expected $\text{Im} / \text{Re Li}_3$ layer	expected $\zeta'(-2)$ flavour

Here Cl_s denotes the Clausen/Glaisher values

$$\text{Cl}_{2k}(\theta) = \text{Im Li}_{2k}(e^{i\theta}), \quad \text{Cl}_{2k+1}(\theta) = \text{Re Li}_{2k+1}(e^{i\theta}),$$

and $\psi^{(-2)}(z) = \int_0^z \log \Gamma(t) dt$ in the Wolfram normalization.

Epistemic status. These are recorded results of experimental mathematics from the PolyLog corpus of the Smithereens repository. Classical statements are attributed; the remaining identities are verified numerically at the stated precision, most of them in *two independent engines* (the Wolfram kernel at 60 digits and `mpmath` at 70 digits), with the project’s PSLQ-canary discipline: every PSLQ discovery must survive re-evaluation at elevated precision, and “independence” claims are supported by negative PSLQ controls at stated coefficient heights (evidence, not proof). The 54-entry polygamma–Clausen store is additionally re-verified in a fresh kernel by the project’s regression suite.

2 The two-way exact DFT

Law 2.1 (forward direction; classical). *For every $n \geq 2$, every $q \geq 1$, and $\gcd(p, q) = 1$,*

$$\text{Li}_n(e^{2\pi ip/q}) = \frac{(-1)^n}{q^n (n-1)!} \sum_{k=1}^{q-1} e^{2\pi i kp/q} \psi^{(n-1)}\left(\frac{k}{q}\right) + \frac{\zeta(n)}{q^n}.$$

This is identity 10.08.03.0007.01 of the `functions.wolfram.com PolyLog` reference [2]; it is provable by rearranging the absolutely convergent Hurwitz-zeta series, and it is the workhorse of the project’s cyclotomic reducer (`cycloExact`): a finite polygamma sum, exact and fast, resolving Li_n at *every* root of unity with no PSLQ, no squarefree-radical basis, and no nested-radical limitation.

Result 2.2 (inverse direction). *Inverting the discrete Fourier transform over $\mathbb{Z}/q$ gives, with $\zeta_q = e^{2\pi i/q}$,*

$$\psi^{(n-1)}\left(\frac{k}{q}\right) = (-1)^n (n-1)! q^{n-1} \sum_{p=0}^{q-1} \zeta_q^{-pk} \text{Li}_n(\zeta_q^p).$$

[verified to 40 digits at weights 2, 3 across $q \leq 21$; re-verified by `mpmath` at 70 dps for $n-1 = 1, 2, 3$ over the full store grid]

The bridge is therefore an *exact finite Fourier transform, in both directions*. Three non-obvious consequences are recorded.

1. **DFT-conjugacy of the relation lattices.** The distribution relation $\sum_j \text{Li}_n(z \zeta_m^j) = m^{1-n} \text{Li}_n(z^m)$ Fourier-transforms into the polygamma multiplication theorem, and polylogarithm inversion/reflection corresponds to the symmetry $k \leftrightarrow q-k$ of the ψ -grid. This is *why* the two relation lattices — computed independently by the project’s two reducers — carry identical dimension counts $\varphi(q)/2$ per denominator grid.

2. **The parity split, and Catalan as the $q = 4$ odd component.** Clausen-type values (Im Li_n at even n , etc.) are the odd part of the ψ -grid under $k \mapsto q - k$; in particular

$$G = \text{Im Li}_2(i) = \frac{\psi^{(1)}(1/4) - \psi^{(1)}(3/4)}{16},$$

which identifies the well-known “arguments 1/4 and 3/4” pairing (Math.SE 893486) as the $q = 4$ odd component of the parity split.

3. **Certificate transport.** Cyclotomic-polylogarithm proof certificates can in principle be transported to and from ψ -side lattice certificates through the exact $\mathbb{Q}(\zeta_q)$ DFT matrix. The ψ -side elementary atoms ($\pi^2 \csc^2$ values) are far lighter than the sine-product algebraic objects that constrain the Γ -certificate sweeps on large grids, so the transport is a proposed escape route from that bottleneck (not yet implemented).

3 Clausen-type laws for polygamma values at rational arguments

MathWorld’s *Polygamma Function* page lists fifteen closed forms in its special-cases block, eqs. (22)–(36) ($\psi^{(1)}$, $\psi^{(2)}$, $\psi^{(3)}$ at p/q with $q \in \{2, 3, 4, 6\}$) in Catalan/Clausen/ β atoms, several of them traditionally presented as isolated curiosities. All of them — and their extension to *every* rational argument — are instances of three sine/cosine-weighted Clausen-DFT laws, obtained by splitting the inverse DFT of Result 2.2 into its elementary (reflection) half and its Clausen half.

Law 3.1 (trigamma; Lewin 1991). *For $0 < p < q$, $\gcd(p, q) = 1$,*

$$\psi^{(1)}\left(\frac{p}{q}\right) = \frac{\pi^2}{2 \sin^2(\pi p/q)} + 2q \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \sin\left(\frac{2\pi j p}{q}\right) \text{Cl}_2\left(\frac{2\pi j}{q}\right).$$

[verified at $q = 3, 4, 5, 8, 12$ to 10^{-46} or better, then over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

This law is classical — MathWorld’s *Trigamma Function* eq. (14) cites Lewin (1991) [1] — and was independently re-derived here from the DFT bridge before the attribution was spotted; the project’s records explicitly re-frame the finding as *placement*, not novelty. MathWorld’s scattered coefficients are its small- q shadows: the $3\sqrt{3}$ of $q = 3$ is $2q \sin(2\pi/3)$, the $8G$ of $q = 4$ is $2q \text{Cl}_2(\pi/2)$.

Law 3.2 (tetragamma).

$$\psi^{(2)}\left(\frac{p}{q}\right) = -\pi^3 \cot\left(\frac{\pi p}{q}\right) \csc^2\left(\frac{\pi p}{q}\right) - 2q^2 \left(\zeta(3) + 2 \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \cos\left(\frac{2\pi j p}{q}\right) \text{Cl}_3\left(\frac{2\pi j}{q}\right) + \varepsilon_q \right),$$

where $\varepsilon_q = (-1)^p \text{Cl}_3(\pi)$ for even q , $\varepsilon_q = 0$ for odd q , and $\text{Cl}_3(\pi) = -\frac{3}{4}\zeta(3)$. [verified over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

Law 3.3 (pentagamma).

$$\psi^{(3)}\left(\frac{p}{q}\right) = \pi^4 \left(2 \cot^2\left(\frac{\pi p}{q}\right) \csc^2\left(\frac{\pi p}{q}\right) + \csc^4\left(\frac{\pi p}{q}\right) \right) + 12q^3 \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \sin\left(\frac{2\pi j p}{q}\right) \text{Cl}_4\left(\frac{2\pi j}{q}\right),$$

equivalently, in difference form, $\psi^{(3)}(p/q) - \psi^{(3)}(1 - p/q) = 24q^3 \sum_j \sin(2\pi j p/q) \text{Cl}_4(2\pi j/q)$. [same verification battery]

The $\psi^{(3)}$ law is the Lewin law one weight up, by the same derivation; no literature citation has been located, and the project records it as “presumably in Lewin’s orbit.” MathWorld’s $162\sqrt{3}$ and $768\beta(4)$ coefficients are $12q^3 \sin(2\pi j/q)$ -instances of the one-sided law ($24q^3$ in the difference form).

Result 3.4 (new instances beyond the MathWorld table). *Among the instances at $q \in \{5, 8, 12\}$:*

$$\psi^{(1)}\left(\frac{1}{5}\right) = \frac{\pi^2}{2 \sin^2(\pi/5)} + 10 \left[\sin \frac{2\pi}{5} \operatorname{Cl}_2\left(\frac{2\pi}{5}\right) + \sin \frac{4\pi}{5} \operatorname{Cl}_2\left(\frac{4\pi}{5}\right) \right],$$

$$\psi^{(1)}\left(\frac{1}{12}\right) = 40 G + \frac{4\pi^2}{(\sqrt{3}-1)^2} + 30\sqrt{3} \operatorname{Cl}_2\left(\frac{2\pi}{3}\right),$$

the latter after duplication/triplication canonicalization collapses the raw five-atom DFT form onto the canonical atom set. Also, MathWorld’s eq. (27) simplifies: from $\operatorname{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$,

$$\psi^{(2)}\left(\frac{1}{3}\right) = -\frac{4\pi^3}{3\sqrt{3}} - 26 \zeta(3).$$

Result 3.5 (the polygamma–Clausen store). *The corpus now carries **identities/polygamma-clausen-values.wl**: 54 entries $\psi^{(m)}(p/q)$ for $m \in \{1, 2, 3\}$, $q \in \{3, 4, 5, 6, 8, 12\}$, all p coprime to q (18 arguments $\times$ 3 orders), in canonicalized Clausen atoms. The residual atom basis across the store is*

$$\begin{aligned} & \{G, \beta(4), \zeta(3)\} \cup \operatorname{Cl}_2\left\{\frac{2\pi}{3}, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{\pi}{4}\right\} \\ & \cup \operatorname{Cl}_3\left\{\frac{2\pi}{5}, \frac{\pi}{4}, \frac{\pi}{6}\right\} \cup \operatorname{Cl}_4\left\{\frac{2\pi}{3}, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{\pi}{4}\right\}. \end{aligned}$$

Every entry is verified in-kernel to $|\text{lhs} - \text{rhs}| < 10^{-50}$ at 60 digits (including a disk round-trip) and independently by an mpmath script at 70 dps (tolerance 10^{-55}) that re-derives the raw DFT bridge, the three laws, all 17 Clausen canonicalization relations, and the L-function bridges of Section 4 from mpmath primitives, with positive and negative PSLQ controls.

4 The boundary of the odd-weight Clausen collapse

A folklore rule says that odd-weight Clausen values at rational angles collapse to rational multiples of zeta values ($\operatorname{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$ being the standard example). The store emission *sharpened the rule’s scope*:

Result 4.1 (crystallographic-angle rule). $\operatorname{Cl}_3(2\pi j/q) \in \mathbb{Q} \cdot \zeta(3)$ holds exactly for lowest-terms denominators with $\varphi(q) \leq 2$, i.e. $q \in \{1, 2, 3, 4, 6\}$ — the crystallographic angles — where only the principal even character survives the cosine DFT. For $q = 5, 8, 12$ the even quadratic character χ_q survives, and $s = 3$ odd against $\chi(-1) = +1$ is the parity-mismatch case, so $\operatorname{Cl}_3(2\pi/5)$, $\operatorname{Cl}_3(\pi/4)$, $\operatorname{Cl}_3(\pi/6)$ are genuine atoms [PSLQ-canaried in mpmath over $\{\zeta(3), \pi^3\}$ at coefficient height 10^{10} , with an independent in-kernel Rationalize canary against $\zeta(3)$], with exact character bridges

$$\begin{aligned} \operatorname{Cl}_3\left(\frac{2\pi}{5}\right) &= -\frac{6}{25}\zeta(3) + \frac{\sqrt{5}}{4}L(3, \chi_5), & \operatorname{Cl}_3\left(\frac{\pi}{4}\right) &= -\frac{3}{256}\zeta(3) + \frac{1}{\sqrt{2}}L(3, \chi_8), \\ \operatorname{Cl}_3\left(\frac{\pi}{6}\right) &= \frac{\zeta(3)}{24} + \frac{\sqrt{3}}{2}L(3, \chi_{12}), \end{aligned}$$

where χ_q is the even quadratic (Kronecker) character mod q .

Consequently the $m = 2$ store entries for $q \in \{5, 8, 12\}$ each carry one genuine Cl_3 atom, while those for $q \in \{3, 4, 6\}$ terminate in $\{\zeta(3), \pi^3\}$ with algebraic coefficients. The negative controls are numerical evidence of independence, not proof; no classical closed form for $L(3, \chi_q)$ in the parity-mismatch case is known.

5 Below the digamma: $\psi^{(-2)}$ at rationals

One integration below the digamma sits $\psi^{(-2)}(z) = \int_0^z \log \Gamma(t) dt$. Integrating Kummer’s Fourier series of $\log \Gamma$ [3] predicts that at rational points this grid lands in $\{\text{Clausen values}\}/\pi \oplus \zeta'(-1) \oplus \text{logs}$: the weight-2 Clausen world enters through the negative-order polygamma ladder, with the Glaisher-layer constant $\zeta'(-1)$ as the genuinely new atom. PSLQ confirms [100-digit canaries; residuals $\leq 10^{-60}$]:

Result 5.1 (log-gamma integrals in the Clausen lattice).

$$\begin{aligned}\int_0^{1/4} \log \Gamma(t) dt &= \frac{3}{32} + \frac{\log 2}{8} + \frac{\log \pi}{8} + \frac{G}{4\pi} - \frac{9}{8} \zeta'(-1), \\ \int_0^{1/3} \log \Gamma(t) dt &= \frac{1}{9} + \frac{\log 2}{6} - \frac{\log 3}{72} + \frac{\log \pi}{6} + \frac{\text{Cl}_2(2\pi/3)}{4\pi} - \frac{4}{3} \zeta'(-1), \\ \int_0^{1/6} \log \Gamma(t) dt &= \frac{5}{72} + \frac{7 \log 2}{72} + \frac{\log 3}{144} + \frac{\log \pi}{12} + \frac{\text{Cl}_2(\pi/3)}{4\pi} - \frac{5}{6} \zeta'(-1).\end{aligned}$$

The structural punchline: the quarter-point integral carries Catalan ($\text{Cl}_2(\pi/2) = G$) and the third-point integral carries $\text{Cl}_2(2\pi/3) = \frac{2}{3} \text{Cl}_2(\pi/3)$, two-thirds of Gieseking’s constant — the “level-3 Catalan” that had surfaced independently in the repository’s Eisenstein-ladder work — now arriving from a completely different direction.

Negative result 5.2 (recorded on purpose). (i) The half-point probe $\int_0^{1/2} \log \Gamma$ initially found no relation — because $\log \Gamma(1/2) = \frac{1}{2} \log \pi$ made the PSLQ basis $\mathbb{Q}$ -dependent, and PSLQ returned that degeneracy instead of the target. Basis atoms must be $\mathbb{Q}$ -independent. (ii) A $\psi^{(-3)}(1/4)$ probe via Cauchy’s repeated-integration kernel produced only a large-coefficient candidate (heights $\sim 10^6$) with a weak canary (10^{-46} at 100 dps) — rejected as a PSLQ false positive. The weight-3 level needs a properly prepared basis (presumably $\zeta'(-2)$ -equivalents plus Barnes- G -layer logarithms).

6 Binet’s second formula and Malmsten’s integrals: the Lambert-kernel face

Binet’s second formula

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t} - 1} dt$$

is the integrated Abel–Plana formula with the *Lambert kernel* $1/(e^{2\pi t} - 1)$ — the same kernel whose moment sums are the Eisenstein q -series of the companion CM article. Malmsten/Vardi log-log integrals $\int_0^1 \ln \ln(1/x) R(x) dx$ are the *Mellin avatars* of Kummer’s Fourier series: expanding the rational kernel R in a geometric series produces $L'(\chi, 1)$ -type sums, and Kummer’s formula converts odd-character $L'(1)$ into $\log \Gamma$ at rationals.

Result 6.1. *Binet’s formula replays exactly at $z = 1, \frac{1}{2}, \frac{1}{4}$ [residuals $\leq 10^{-52}$], turning every Γ -store atom into a closed-form arctan–Lambert integral; e.g.*

$$\int_0^\infty \frac{\arctan(4t)}{e^{2\pi t} - 1} dt = \frac{1}{2} \left[\log \Gamma\left(\frac{1}{4}\right) - \frac{1}{2} \log 2 + \frac{1}{4} - \frac{1}{2} \log(2\pi) \right].$$

Result 6.2 (Vardi’s integral [5], re-derived blind). *PSLQ over $\{\pi\gamma, \pi \log 2, \pi \log \pi, \pi \log \Gamma(1/4)\}$, with the γ atom declined (coefficient 0) [canary exact at 80 dps]:*

$$\int_0^1 \frac{\ln \ln(1/x)}{1+x^2} dx = \frac{\pi}{4} \ln\left(\frac{4\pi^3}{\Gamma(1/4)^4}\right).$$

The $q = 3$ Malmsten variant [canary 3.4×10^{-81} ; γ declined again]:

$$\int_0^1 \frac{\ln \ln(1/x)}{1+x+x^2} dx = \frac{\pi}{6\sqrt{3}} \ln\left(\frac{2^8 \pi^8}{3^3 \Gamma(1/3)^{12}}\right).$$

Both integrals carry exactly the harvested- Γ -store atoms: the log-log family is a systematic *integral-representation generator* for the identity stores — every odd Dirichlet character (each denominator q) yields one such evaluation through Kummer’s formula, with the Γ -lattice reducer supplying the closed form mechanically.

7 The log-gamma moment ladder

Result 7.1 (Hurwitz ladder; classical (Whittaker–Watson [8])).

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{2} \sum_{n \geq 2} \frac{n-1}{n(n+1)} \zeta(n, z+1) \quad [\text{verified exactly at } z = 1/3].$$

Since $\zeta(n, z+1)$ is $\psi^{(n-1)}(z+1)$ up to factorials, this is a $\mathbb{Q}$ -linear ladder between $\log \Gamma$ and the entire polygamma tower at the same point — a candidate certificate law binding the Γ and trigamma stores into one linear system.

Result 7.2 (second moment; classical (Espinosa–Moll [7])). *With $L = \log(2\pi)$,*

$$\int_0^1 [\log \Gamma(x)]^2 dx = \frac{\gamma^2}{12} + \frac{\pi^2}{48} + \frac{\gamma L}{6} + \frac{L^2}{3} - (\gamma + L) \frac{\zeta'(2)}{\pi^2} + \frac{\zeta''(2)}{2\pi^2} \quad [\text{verified to } 1.6 \times 10^{-61}].$$

The quadratic-moment layer adds $\zeta''(2)$ to the atom inventory.

Negative result 7.3 (the cubic moment stays open).

$$\int_0^1 [\log \Gamma(x)]^3 dx = 5.7403888072294742800195716881024614629610130074549 \dots$$

(50 digits, tanh-sinh quadrature cross-calibrated by the second moment). PSLQ over the natural weight-mixed basket $(\gamma^3, \dots, L^3, \pi^2\gamma, \pi^2L, \zeta(3), (\gamma+L)\zeta'(2)/\pi^2, (\gamma+L)\zeta''(2)/\pi^2, \zeta'''(2)/\pi^2, \zeta'(2)^2/\pi^4, \zeta'(3), \gamma\zeta'(2)/\pi^2)$ returns only uniform-height junk at 60 dps — consistent with Espinosa–Moll and Bailey et al. The Kummer-cube resonance $k_1 \pm k_2 \pm k_3 = 0$ produces a genuinely new depth-2 constant (Tornheim–Witten-derivative flavour); finding the right named atom is the open problem, not search depth.

8 The Stieltjes layer

Differentiating $L(s, \chi) = q^{-s} \sum_k \chi(k) \zeta(s, k/q)$ at $s = 1$ gives the exact ladder

$$L'(\chi, 1) = -\ln q L(1, \chi) - \frac{1}{q} \sum_k \chi(k) \gamma_1\left(\frac{k}{q}\right),$$

so every Malmsten log-log integral is a statement about *first generalized Stieltjes constants* $\gamma_1(k/q)$ — Blagouchine’s program, equivalent to Deninger’s $\zeta''(0, a)$ layer via the functional equation.

Result 8.1 (even-character Malmsten integral). *[verified to 10^{-51} ; even character mod 5; φ the golden ratio]*

$$\int_0^1 \ln \ln \frac{1}{x} \cdot \frac{(1-x)(1-x^2)}{1-x^5} dx = -(\gamma + \ln 5) \frac{2 \ln \varphi}{\sqrt{5}} - \frac{\gamma_1(\frac{1}{5}) - \gamma_1(\frac{2}{5}) - \gamma_1(\frac{3}{5}) + \gamma_1(\frac{4}{5})}{5}.$$

Result 8.2 (the parity dichotomy). *The dichotomy is empirically sharp. The odd γ_1 -combination reduces exactly as differentiated-Kummer predicts [verified blind]:*

$$\gamma_1(\frac{1}{3}) - \gamma_1(\frac{2}{3}) = -\frac{\pi}{\sqrt{3}} \left(\gamma + 4 \ln 2 - \frac{1}{2} \ln 3 + 4 \ln \pi - 6 \ln \Gamma\left(\frac{1}{3}\right) \right),$$

which is Malmsten’s 1846 reflection identity [4] (long misattributed to Almkvist–Meurman). The even combination does not reduce to standard atoms [PSLQ junk at coefficient height 10^8 — honest negative]; pushing Blagouchine’s rational-arguments theorem [6] through it collapses everything except a ζ'' -block [derived and verified to 10^{-50}]:

$$\gamma_1(\frac{1}{5}) - \gamma_1(\frac{2}{5}) - \gamma_1(\frac{3}{5}) + \gamma_1(\frac{4}{5}) = \sqrt{5} \left[\zeta''(0, \frac{1}{5}) - \zeta''(0, \frac{2}{5}) - \zeta''(0, \frac{3}{5}) + \zeta''(0, \frac{4}{5}) \right] - 2\sqrt{5} \ln \varphi (\gamma + \ln 10\pi).$$

The canonical atom for the even layer is thus the even-character $\zeta''(0, l/q)$ combination (Deninger’s layer [9]); Malmsten integrals, even γ_1 -combinations, and even $L'(\chi, 1)$ are all elementary-equivalent to it. Two practical records: Blagouchine’s particular value $\gamma_1(1/4)$ was re-verified exactly against `mpmath`’s generalized Stieltjes constants, and the reduction coefficients live in $\mathbb{Q}(\sqrt{q})$ — rational-coefficient PSLQ baskets *miss* the relation, so a $\sqrt{q}$ -weighted basket or direct derivation is required.

9 Open directions

1. Attribution: a literature citation for the $\psi^{(3)}$ law (Law 3.3) and the general- m odd-half Clausen-DFT version; only the trigamma law’s Lewin 1991 attribution is confirmed.
2. Implementing certificate transport across the exact $\mathbb{Q}(\zeta_q)$ DFT matrix (Section 2, consequence 3).
3. Closed forms (or proofs of independence) for $L(3, \chi_5)$, $L(3, \chi_8)$, $L(3, \chi_{12})$ — equivalently for the three genuine Cl_3 atoms.
4. The $\psi^{(-3)}$ layer with a properly prepared basis; a store schema for integral/infinite-sum claims; the named atom for the cubic log-gamma moment.
5. Extending the polygamma–Clausen store beyond $m \leq 3$, $q \in \{3, 4, 5, 6, 8, 12\}$, and equipping its entries with replayable proof certificates.

Sources and verification artifacts

This article presents results recorded in the Smithereens repository (`src/PolyLog`) in the commit range `272af472..68236dab4` (2026-06-09 through 2026-06-11): the two-way DFT in `docs/reduction-method.md` §4; the Clausen laws and store in `identities/polygamma-clausen-values.wl` with generator `tools/generate-polygamma-clausen-store.wl` and cross-engine verifier `tools/verify-polygamma-clausen-mpmath.py`; the $\psi^{(-2)}$ bridge in `docs/reports/loggamma-a-integrals-clausen-bridge__c5013328db83.md`; the Binet/Malmsten/Stieltjes material in `docs/reports/binet-malmsten-lambert-bridge__a3c91e07f5d2.md`.

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